

RECURSION QUESTIONS

Question 1 : For a given integer array of size N. You have to find all the occurrences (indices) of a given element (Key) and print them. Use a recursive function to solve this problem.

Sample Input : arr[] = {3, 2, 4, 5, 6, 2, 7, 2, 2}, key = 2

Sample Output : 1 5 7 8

Question 2 :

You are given a number (eg - 2019), convert it into a String of english like "two zero one nine". Use a recursive function to solve this problem.

NOTE- The digits of the number will only be in the range 0-9 and the last digit of a number can't be 0.

Sample Input : 1947

Sample Output : "one nine four seven"

Question 3 : Write a program to find Length of a String using Recursion.

Question 4 : We are given a string S, we need to find the count of all contiguous substrings starting and ending with the same character.

Sample Input 1 : S ="abcb"

Sample Output 1 : 7

There are 15 substrings of "abcb": a, ab, abc, abca, abcb, b, bc, bca, bcab, c, ca, cab, a, ab, b
Out of the above substrings, there are 7 substrings : a, abca, b, bcab, c, a and b. So, only 7 contiguous substrings start and end with the same character.

Sample Input 2 : S ="aba"

Sample Output 2 : 4

The substrings are a, b, a and aba.